Exam Assignment

Machine Learning (BSc Data Science) Fall 2026

IT University of Copenhagen

1 Introduction and formalities

This is the project description for the exam project in the Machine Learning course for the BSc program in Data Science at the IT University of Copenhagen. The project must be submitted electronically via Wiseflow no later than **December 18, 2 pm**.

Groups. Please register as a group on Wiseflow by the end of the day on October 23. The project is designed for groups of 3 students. Groups of 2 or 3 people are allowed. If you have any questions regarding the group formation, please talk to me (Jonas, the course manager) before the group formation deadline.

I reserve the right to modify the grouping if necessary. If you need help finding a group, get in touch with me before the group formation deadline. Any students who have not registered as part of a group will be allocated to a group (possibly an existing one) at random.

Only one person per group should submit the project.

1.1 Partnership with Grundfos

The Machine Learning project investigates the flow of water through pumps, and we are happy to announce that the project has been designed in partnership with a performance and data science team at the world-leading Danish pump company Grundfos. The data are simulated with the company's own models, and the questions you will pursue are similar to those a company data scientist might encounter.

The project is a normal university project in that it is evaluated by the course manager, according to criteria laid out in the course description.

The project has a special addition, though: I will pick a couple of particularly good submissions, and share them with representatives from Grundfos's data science team. From these nominated projects, they will pick a winning project. The authors of the winning project will receive a prize.

In case you do not consent to us sharing your project with the external partners at Grundfos (which we will only do if we deem your project to be among the very best ones), you must opt out of the competition when you submit the project via Wiseflow (opt-out happens in the submission portal).

1.2 What should be handed in?

You must hand in both a report and the source code you have developed during the project compressed as a single `.zip` file. Note that the project's evaluation will be based almost entirely on the report; the source code should be seen as a supporting document.

Make sure to use correct references in your report, including references to any of the course textbooks. This also applies to code; if you copy or take inspiration from code developed by others, this should be stated clearly in your report. This goes for **Generative Artificial Intelligence** (GenAI), too. You can use GenAI as an aid in developing code, but do not copy-paste or accept suggestions without scrutinizing and editing carefully, do make sure to give references, and remember that *you* are the expert. Please read and comply with the university's guide on using GenAI[1]. Note that any work based on a previous exam submission for this course, even if it is your own work, needs to be clearly marked as such and cited.

[1] `https://itustudent.itu.dk/Study-Administration/Generative-AI`

Report. The report should be submitted as a single PDF file. There is a strict limit of 10 pages, including figures, tables, code snippets, references, and appendices, but excluding the front page. The report must be typeset with at least 11pt font size and margins of at least 2 cms. The report must be in PDF format and have a front page that meets the ITU requirements.[2]

Implementation and code. Your implementation has to be in Python. Except where explicitly stated, there are no restrictions as to which Python libraries you may use.

Your code should be organized such that it is easy to read, i.e. you have to use descriptive names for files, functions, variables, etc. The code may be organized in regular Python source files (`.py` files) or Jupyter notebooks (`.ipynb` files) and either submitted as part of the project or uploaded to a repository that I have access to.

2 Problem and data set

The project datasets include measurements from multiple pumps, covering various pump and flow properties along with the corresponding flow rate at the time of each measurement. Figure 1 shows an illustration of a pump in action. We consider two cases. In `case 1`, the flow is always stable and the same pump models appear in both the training and test data. `Case 2` is more challenging: the flow is sometimes transient, and no pump appears in both datasets. The table below lists and describes the variables in the datasets; some are present only in `case 2`.

Table 1: Variables in the pump datasets.

Name	Symbol	Unit	Description
`pump_id`	—	—	Identifier of the pump unit.
`elapsed_s`	t	s	Time since the start of the recording for each pump, sampled once per second.
`operating_state`	—	—	Operating regime label: *stable* (steady state) or *transient* (changing operating point).
`speed_rpm`	n	rpm	Rotational speed of the pump shaft.
`power_w`	P	W	Shaft power delivered to the pump, equal to torque times angular speed.
`inlet_pressure_bar`	p_{in}	bar	Static pressure on the suction side of the pump.
`outlet_pressure_bar`	p_{out}	bar	Static pressure on the discharge side of the pump.
`measured_head_m`	H	m	Pump head, $H = \Delta p/(\rho g)$; the pressure rise expressed as a height of water.
`current_a`	I	A	Electrical current drawn by the motor driving the pump.
`torque_nm`	τ	N m	Torque applied to the pump shaft by the motor.
`temperature_c`	T	°C	Temperature of the pumped fluid.
`flow_m3_h`	Q	$\text{m}^3\,\text{h}^{-1}$	Volumetric flow rate through the pump; **the prediction target.**
(derived)	Δp	bar	Pressure rise across the pump, $\Delta p = p_{\text{out}} - p_{\text{in}}$.

The datasets representing `case 1` and `case 2` are each divided into a training set and a test set, available in `.csv` format as:

`Case1:` `train1.csv` and `test1.csv`
`Case2:` `train2.csv` and `test2.csv`

3 Scientific requirements to the project

Carry out all of the tasks listed below and report on your findings.

[2]Found at: `https://itustudent.itu.dk/Study-Administration/Projects/Project-exams/Submitting-written-work-projects`

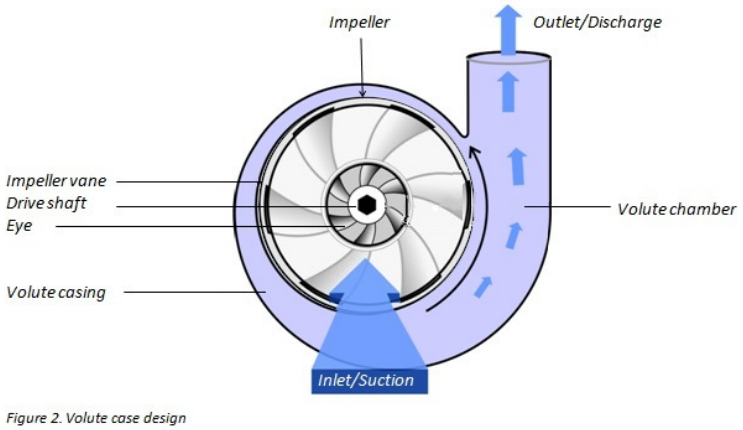


Figure 1: Different parts of a pump. Source: `https://www.michael-smith-engineers.co.uk/resources/useful-info/centrifugal-pumps`

3.1 Task 0: Recovering pump physics

Before you turn to the project's main task of predicting pump flow, we ask you to approximate a central equation from pump physics using machine learning.

In pump physics, the power in a pump P can be expressed as a function of the variables n (kilo rotations per minute) and Q (the flow created by the pump in m^3/h). The formula is a polynomial:

$$P = \sum_{i=0}^{3} \sum_{j=0}^{3} \alpha_{ij} n^i Q^j, \tag{1}$$

where α_{ij} are real-valued coefficients, but because you are IT experts and not engineers, you never learned exactly what the expression is (that is, which combinations of i and j are part of the formula, and which are left out).

Task 0: Use Linear Regression to predict P from features that combine the variables n and Q as shown in (1).

Recover the regression expression that best describes P. Use `train1.csv` as your training dataset and report your prediction error as evaluated on `test1.csv`.

Subtasks:

- *Create relevant new feature columns by combining existing features. (remember to convert n from rpm (rotations per minute) to krpm)*
- *Fit Linear Regression models with relevant new and old feature columns.*
- *Apply some feature selection method to find the polynomial of form* (1) *that matches the data best.*
- *Report your final regression expression and your test errors, using one or more informative error measures of your choice, and show a plot of errors versus flow.*

3.2 Predicting flow (Case 1)

For a company that manufactures and monitors pumps, the variables n, P, and Δp can be read off directly from the pump they deliver to the customer. What is hard to know is the flow of water Q at the site. Therefore, it is important to design machine learning models that accurately predict Q from combinations of the other three variables.

We expect the prediction task to be nonlinear. Nevertheless, we first construct a linear baseline model against which to compare more advanced models.

Task 1: Use Linear Regression to predict Q from the variables n, P, Δp, and any combinations thereof that you find relevant. Use the dataset `train1.csv` as your training data, and `test1.csv` as your test data. Report your test error and the parameters of your best-performing regression model.

Task 2: Use other regression models to predict Q from the variables n, P, Δp, and any combinations thereof that you find relevant. Use the dataset `train1.csv` as your training data, and `test1.csv` as your test data. Report your test errors and the hyperparameters (and how you found them) of your best-performing regression model. You should explore at least 3 Machine Learning models:

M1 Decision Tree Regressor

M2 Feed-forward Neural Network Regressor

M3 One or more other methods of your own choice, not limited to regression methods.

M1 and M2, should be implemented in two versions:

1. An implementation from scratch (training and prediction) using only Python standard libraries and the numerical libraries NumPy and SciPy.

   Thus, you cannot use machine learning libraries such as TensorFlow, PyTorch, Scikit-learn, or Keras. Note that the restriction only applies to the implementation of the method itself (training and prediction), but not to any pre-processing before you feed the input to the method or the further interpretations of the results, such as visualizations.

2. A "reference implementation" using any Python library which can help you assess the correctness of your own implementation.

3.3 Predicting flow (Case 2)

`Case 2` mimics a situation where the company is using in-house pumps to understand the behavior of pumps that are located at customer sites. The training set (`train2.csv`) has data derived from 5 pumps and the test set (`test2.csv`) contains data from 10 other pumps.

Task 3: Predict flow using the features available in the `case 2` data sets (and any combinations of these features that you deem relevant). Choose any machine learning model you wish (noted as M4), and try to get the most accurate prediction possible. Report the test errors of your best-performing model and argue for your choice of performance metrics. You should also include a flow versus error plot in your results. Additionally, make sure to discuss the hyperparameters of your model, including any relevant training and hyperparameter selection details.

Suggestion 1: Drop the column `elapsed_s` as a feature.

Suggestion 2: Since data are derived from different pumps, any splitting of the data should be done carefully in order to avoid data leakage. If you want to hold out some data as part of your training process, hold out all data points of a pump simultaneously. I suggest using `scikit-learn`'s `GroupKFold` for this.

3.4 Report

The report should contain the following things.

Data Cleaning and Exploratory Data Analysis. Your report should introduce the reader to the data by illustrating selected aspects of the data, and argue which features needed cleaning and explain why.

Description of feature extraction and selection. If you choose to coarse-grain certain features, combine features, disregard features, etc., your report should explain how you do this and why.

Visualization of data. As part of the exploratory data analysis, your report should present a visualization of the dataset in any way you see fit. Remember to state what the reader can learn from the visualization.

Details on implementations. Your report should describe and discuss the key points of how you implemented methods M1, M2, M3, and M4. For each method please make sure to include:

- A description of how you applied the method to the data, including details needed for an independent reproduction of your results.
- A discussion on how you have gone about selecting any hyperparameters for the method.
- A discussion on how you have assessed your implementation's correctness (i.e. did it do as intended?). Do note that correctness and performance are not interchangeable terms.

Interpretation and discussion of the results. Your report should include a thorough discussion of the comparative performance of methods applied. In particular, you should compare each method's performance and guide the reader in interpreting the results. Use your expert knowledge to explain the results; for instance, why do particular methods perform better or worse than others?

Outlook. A company like Grundfos can alter their data collection, the monitoring setup, etc. to improve their predictions. What would you do to improve the prediction? Your report should comment on this problem and I would be curious to hear any ideas you may have.

4 Final comment

Modeling and predicting flow in realistic settings is a hard, but important problem. Your job is to do your best to improve your model's performance, and to make informed and reasonable decisions at every step of your modeling journey.

Have fun working on the project and I look forward to seeing your reports.